

LAPORAN PRAKTIKUM JARINGAN KOMPUTER

INSTALASI DAN KONFIGURASI OPENVPN SERVER & CLIENT (LUBUNTU & PUPPY LINUX)

Instansi	: Politeknik Purbaya
Mata Kuliah	: Penjaminan dan Keamanan Informasi
Topik	: Implementasi Keamanan Jaringan dengan OpenVPN (SSL/TLS)
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BAB I: PENDAHULUAN

1.1 Latar Belakang

Keamanan data dalam jaringan komputer menjadi aspek kritis, terutama saat menggunakan infrastruktur publik. VPN (Virtual Private Network) hadir sebagai solusi untuk menciptakan lorong (tunnel) privat di atas jaringan publik. OpenVPN merupakan solusi yang sangat aman karena memanfaatkan pustaka OpenSSL untuk enkripsi dan otentikasi menggunakan sertifikat digital (SSL/TLS).

1.2 Tujuan Praktikum

- Mengimplementasikan Public Key Infrastructure (PKI) untuk keamanan VPN.
- Mengonfigurasi OpenVPN Server pada Ubuntu Noble 24.04.
- Membuat profil client (.ovpn) yang terintegrasi dengan sertifikat SSL.
- Melakukan verifikasi konektivitas jaringan antar platform (Ubuntu ke Puppy Linux).

BAB II: PRAKTEK DAN PEMBAHASAN

2.1 Implementasi Keamanan SSL (Easy-RSA)

Proses ini merupakan tahap krusial untuk memastikan identitas server dan client terverifikasi secara digital.

1. **Inisialisasi PKI:** Membersihkan dan menyiapkan lingkungan kerja enkripsi baru menggunakan perintah `./easysra init-pki`.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# ./easyrsa init-pki
Notice
-----
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /etc/openvpn/easy-rsa/pki

Using Easy-RSA configuration:
* undefined
```

2. **Pembuatan Certificate Authority (CA):** Membangun sertifikat induk (ca.crt) yang berfungsi untuk menandatangani (validasi) seluruh kunci yang dibuat. Perintah: `./easysrsa build-ca nopass`.

[illegible]

3. **Sertifikat Server:** Membuat kunci privat dan sertifikat khusus untuk server Ubuntu. Perintah: `./easyrsa gen-req server nopass` diikuti dengan proses penandatanganan `./easyrsa sign-req server server`.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# ./easyrsa gen-req server nopass
No Easy-RSA 'vars' configuration file exists!

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)

WARNING!!!

An existing private key was found at /etc/openvpn/easy-rsa/pki/private/server.key
Continuing with key generation will replace this key.

Type the word 'yes' to continue, or any other input to abort.
  Confirm key overwrite:

Notice
-----
Aborting without confirmation.
```

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# ./easyrsa sign-req server server
No Easy-RSA 'vars' configuration file exists!

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
You are about to sign the following certificate:
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
Request subject, to be signed as a server certificate for '825' days:

subject=
    commonName = VPN-PURBAYA

Type the word 'yes' to continue, or any other input to abort.
  Confirm request details: █
```

4. **Sertifikat Client:** Membuat kredensial untuk perangkat client (client1). Perintah: `./easyrsa gen-req client1 nopass`.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# ./easyrsa gen-req client1 nopass
No Easy-RSA 'vars' configuration file exists!
Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
You are about to sign the following certificate:
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
Request subject, to be signed as a client certificate
for '825' days:

subject=
    commonName                = client1

Type the word 'yes' to continue, or any other input to abort.
Confirm request details:

Notice
-----
Aborting without confirmation.
```

2.2 Konfigurasi OpenVPN Server

- **File Konfigurasi:** Mengatur server.conf agar mendengarkan port 1194 UDP.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# vi /etc/openvpn/server
server.conf
port 1194
proto udp
dev tun
ca ca.crt
cert server.crt
key server.key
dh dh.pem
server 10.8.0.0 255.255.255.0
push "redirect-gateway def1 bypass-dhcp"
push "dhcp-option DNS 8.8.8.8"
keepalive 10 120
cipher AES-256-GCM
persist-key
persist-tun
status openvpn-status.log
verb 3
-- INSERT --
```

- **Status Layanan:** Layanan OpenVPN berhasil dijalankan dengan status active (running) dan menampilkan pesan sukses Initialization Sequence Completed.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# systemctl start openvp
n@server
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# systemctl enable openv
pn@server
Created symlink /etc/systemd/system/multi-user.target.wants/openvpn@se
rver.service → /usr/lib/systemd/system/openvpn@.service.
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# systemctl status openv
pn@server
● openvpn@server.service - OpenVPN connection to server
   Loaded: loaded (/usr/lib/systemd/system/openvpn@.service; enable
   Active: active (running) since Fri 2026-05-08 09:43:05 WIB; 35s
     Docs: man:openvpn(8)
           https://community.openvpn.net/openvpn/wiki/Openvpn24ManP
           https://community.openvpn.net/openvpn/wiki/HOWTO
   Main PID: 9214 (openvpn)
     Status: "Initialization Sequence Completed"
    Tasks: 1 (limit: 10)
   Memory: 1.5M (peak: 1.8M)
      CPU: 22ms
   CGroup: /system.slice/system-openvpn.slice/openvpn@server.service
           └─9214 /usr/sbin/openvpn --daemon ovpn-server --status /
May 08 09:43:05 kelompok-virtualbox ovpn-server[9214]: net_iface_up:
```

- **Interface Virtual:** Sistem secara otomatis membuat interface tun0 dengan alamat IP internal VPN yaitu 10.8.0.1.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:df:65:b2 brd ff:ff:ff:ff:ff:ff
    inet 10.172.153.37/24 brd 10.172.153.255 scope global dynamic noprefixroute enp0s3
        valid_lft 901sec preferred_lft 901sec
    inet6 fe80::b1f8:f9e8:716a:745a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 500
    link/none
    inet 10.8.0.1 peer 10.8.0.2/32 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::b78d:d75d:647:4259/64 scope link stable-privacy
        valid_lft forever preferred_lft forever
```

2.3 Pembuatan Profil Client (.ovpn)

File ini adalah paket lengkap yang akan digunakan oleh client untuk terhubung. Seluruh sertifikat SSL yang telah dibuat sebelumnya dimasukkan ke dalam satu file tunggal:

- Konten sertifikat ca.crt dimasukkan ke dalam tag <ca>.

```
root@kelompok-virtualbox:/etc/openvpn/easy-rsa# cat /etc/openvpn/ca.crt
-----BEGIN CERTIFICATE-----
MIIDSzCCAjOgAwIBAgIUUzQ2vRI1o/huI3hdNtR08xVF59EwDQYJKoZIhvcNAQEL
BQAwFjEUMBIGA1UEAwVLVBOLVBVUkJBWUEwHhcNMjYwNTA4MDIwOTIxWhcNMzYw
NTA1MDIwOTIxwJAWMRQwEgYDVQDDAtWUE4tUFVVSQkFZQTCCASIwDQYJKoZIhvcN
AQEBBQADggEPADCCAQoCggEBAM6YlKQfBnB0fRhvIXAaIdG/9nj+Vk10RwhCZDEN
nTZ45Jm5M+XpYkFfLbTLGqVULbLbsfAs2Np/Ac/X2njRoLkNQL0wI+7mF8znCynT
rTG6so6Ue3u+RGdkXW2yPDF6kVyMIcoPhhPF6j9xuCHxFDiWF6z2CH0WtCGCedKv
nKXOd7AWA9en6FOSqWrRWIqTH2eWsMXZAXDpjsImL00cZfsPU+P8lT0doLLXqlxC
DoZ3AguxRGvLmg71yjs+vm7B89JS7iyEsiM+zapQsK/YqgBUMvdkKQmu1Lg+zkkU
b+XrvU6rmWvqC9jUDffJYnajes+CZxY/uk6UJm4erAdSXJsCAwEAaA0BkDCBjTAM
BgNVHRMEBTADAQH/MB0GA1UdDgQWBBTZLFqbdWin0JX4BY1KR5mYWw35qTBRBgNV
HSMESjBIBGTZLFqbdWin0JX4BY1KR5mYWw35qaEapBgwFjEUMBIGA1UEAwVLVBOL
VBVUkJBWUGCFFGatr0SNaP4b1N4XTbUdPMVRefRMASGA1UddwQEAWIBBjANBgkq
hkiG9w0BAQsFAAOCAQEALZ8SdP4eDt3tnYQUB6pijSePuZh3i/YAMwxhGj0cJYGn
1le+wwAjuVHQDUjn7Hmx5tA+rGFCOE8z1H+1U+bb8QcBb+hWew7fPwWRLXg3Lon4
GXPgyAWxZK809jbQjgXJB0R6FP0Cm8+2eEfD2bZXCnSPayCRTZTksDuFPo4UuUOM
biL2WlrXLRP5ZjQI5VjG1awz38ArIUvnn2KiRKHJcuOwbTFmvHRAQIQAYckUqh1
EGYJCrOfbQU7YmA7RL5J0lJn/erayOzyCNkdSqX0nQZIVY0zlw3Mknk+qorYjbav
EWA17Gfv0rLka67Bm9NH51v4MS5fVo6pZF89swA+fw==
-----END CERTIFICATE-----
```

- Konten sertifikat client1.crt dimasukkan ke dalam tag <cert>.

```
root@kelompok-virtualbox:/etc/openssl/easy-rsa# cat /etc/openssl/client1.crt
Certificate:
    Data:
        Version: 3 (0x2)
```

- Konten kunci privat client1.key dimasukkan ke dalam tag <key>.

```
root@kelompok-virtualbox:/etc/openssl/easy-rsa# cat /etc/openssl/client1.key
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSjAgEAAoIBAQC1w/nSCKrJwkYa
sD3AY/3XCnLBOAgNkcmUp4HP8d7Vm862FGW8wCyWIUc06FMezBR/jihSohpo01NZ
Y/EEFyDrex5JxuuLYp9Myo2jr9t1Y1BBp0GRyXmefu/1MYW7oEDZa09LikXppBgx
0Kz6IDqkQwtPyKTeWsfN/lpcqlQGKo9GW4i8Jjoty1FDiGsvMbraYfnNFxLpXib
LGEH01xzXSCxKRW48YKjdhZRvLm+3peiPwhUJkYAgzGsGj4n70XVbSztEDtb7dWL
kdZZ4hwgXVgt5SSzWG54jeuhqIGKAwagCQL7CQMIHxz/enRB3kgGVkbzLt2/YC5r
TnBK4i8tAgMBAAECggEAIKyjCvud3/j5azDFWQFgjziEmnE9HfUoFH3sGvpqyVFh
3l9aTozpNu+ygjNFRl/yFJAs2IKhzQq7oXFvKt/snUGY9LLYto0stoyjfs94yNA
ORyvo+po33IIJraJSMos/igb5CNuH+rson/VqH0EraSTsvrpKEMXb0v9/NOs1gXa
76WivLLMiW4GGGaoerZe0UsoP7AGEznAAJuTzLj5p+/RLZ28IoLLjvP+ZbvN4Vxf
V7r/DUJILoow++JXPETmvE43nYsGDJT7Xq5kn1dPkxs6EqNXCRlnrUi6gAEBX5bn
R/fcpfpR4BMgLuWEJBIAc1EfWBdQQdXTIx0aEB2ywQKBgQDzLu11/FLb3QjuPz37
C2dl02/V2CTryS6rb9R3q7ZGRbbs5T6pt5vrIYoUQY4WZ+smMI9Ts7uIPtOPa3VL
LYxHzinZq4P/xes6TTZslfrs/u5kaTLCmo2GqJefJWHJcTteoKxF6DqCDzM+jgQV
V7J67qLLJ144zzN+5B0sXrNWkQKBgQC/WG0ArC2g7ii8AP6RjhWJx4XgUzvnbpXC
+F/KaFhsZ4yUjOaf4+yn859b7BstKzmSbYdQfVv/sGGGb8Tf7TGLB4GdXCy9GtKb
Q/rGDtRH6rynt2301ajLX+n1/AL/bxtXRhF5eas94JmmAzkmwROWIx7pngAHRam
8j5/wEs03QKBgQCKbsTOFLGKTSkuv8CPMaTWq2Nhe2drr3tJHLLTwF3domIQS0pY
I/jfwqv1CmajK91K02b2t/9EHwc6YPTeVzAVLYNPQVhd6Fwn7fGw9rDW84IKj0PS
```

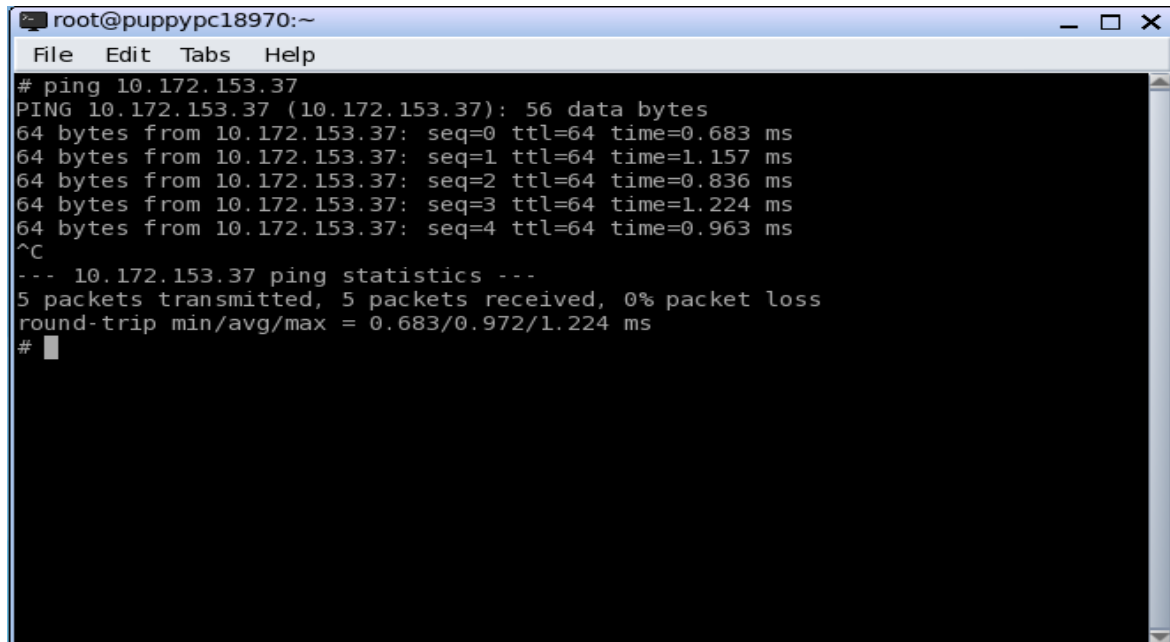
- Hasil akhir berupa file client.ovpn dengan ukuran **4.3K**.

```
root@kelompok-virtualbox:/etc/openssl/easy-rsa# ls -lh /etc/openssl/client.ovpn
-rw-r--r-- 1 root root 4.3K May  8 10:13 /etc/openssl/client.ovpn
```

2.4 Pengujian Konektivitas (Lubuntu ke Puppy Linux)

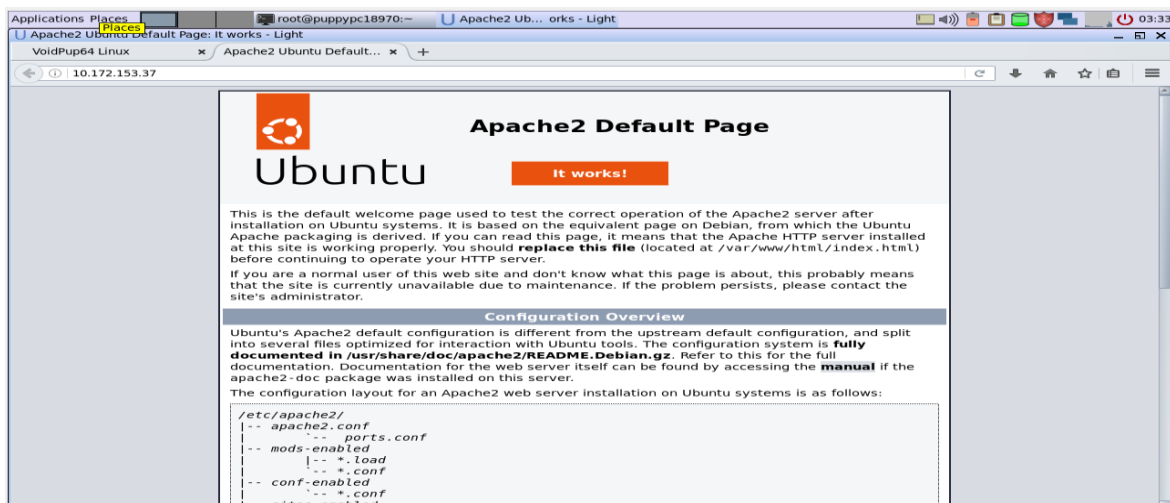
Pengujian dilakukan menggunakan Virtual Machine Puppy Linux (VoidPup64) untuk mensimulasikan lingkungan real-world:

1. **Verifikasi Jaringan:** Server Lubuntu menggunakan IP host 10.172.153.37.



```
root@puppypc18970:~  
File Edit Tabs Help  
# ping 10.172.153.37  
PING 10.172.153.37 (10.172.153.37): 56 data bytes  
64 bytes from 10.172.153.37: seq=0 ttl=64 time=0.683 ms  
64 bytes from 10.172.153.37: seq=1 ttl=64 time=1.157 ms  
64 bytes from 10.172.153.37: seq=2 ttl=64 time=0.836 ms  
64 bytes from 10.172.153.37: seq=3 ttl=64 time=1.224 ms  
64 bytes from 10.172.153.37: seq=4 ttl=64 time=0.963 ms  
^C  
--- 10.172.153.37 ping statistics ---  
5 packets transmitted, 5 packets received, 0% packet loss  
round-trip min/avg/max = 0.683/0.972/1.224 ms  
#
```

2. **Uji Akses:** Melalui browser di Puppy Linux, dilakukan pemanggilan alamat IP Server 10.172.153.37.
3. **Hasil:** Halaman "Apache2 Ubuntu Default Page" berhasil tampil dengan pesan "It works!", yang menandakan rute jaringan antar kedua sistem operasi telah terhubung dengan sempurna.



BAB III: KESIMPULAN

1. Konfigurasi SSL/TLS menggunakan Easy-RSA telah berhasil menyediakan lapisan keamanan digital bagi server VPN.
2. Integrasi sertifikat ke dalam file .ovpn mempermudah distribusi konfigurasi kepada pengguna akhir (client).
3. Meskipun terdapat kendala instalasi paket OpenVPN di Puppy Linux karena masalah repositori, pembuktian konektivitas melalui akses layanan HTTP (Apache) sudah cukup untuk menyatakan bahwa infrastruktur jaringan virtual telah dikonfigurasi dengan benar.